

MambaRefine-CD: Results Summary

Table 1: Comparison on DSIFN-CD.

Model	Pre	Rec	F1	IoU	OA
FC-Siam-Conc	66.45	54.21	59.71	42.56	87.57
SNUNet	60.60	72.89	66.18	49.45	87.34
ChangeFormer	88.48	84.94	86.67	76.48	95.56
BiFA	73.99	68.87	71.34	55.45	90.80
FTAN	90.54	88.61	89.56	81.10	—
ADSFNet	<i>94.79</i>	95.24	<u>95.01</u>	90.50	98.30
Mamba-CD	95.60	<u>95.61</u>	<i>95.61</i>	<i>91.69</i>	98.51
MambaRefine-CD (ours)	<u>95.47</u>	95.87	95.67	91.71	<i>96.98</i>

Table 2: Ablation study on DSIFN-CD clean split. F1, IoU, and OA are reported for the changed class.

ID	Variant	Main Added Com- ponent	Params (M)	F1	IoU	OA
A0	FPN Baseline	SimpleCNN encoder	7.84	76.63	62.12	83.80
A1	MambaVision-S + FPN	MambaVision en- coder	53.54	93.21	87.28	95.72
A2	+ D-RBI unsigned	Absolute-diff D-RBI	54.98	93.33	87.50	95.35
A3	+ Signed Diff	Signed temporal stream	55.34	94.28	89.19	95.99
A4	+ ARF-FPN	Adaptive RF decoder	65.12	94.36	89.32	96.05
A5	+ Boundary Residual	Bounded residual head	65.19	93.59	87.94	95.53

Table 3: Comparison on WHU-CD.

Method	Pre	Rec	F1	IoU	OA
FC-EF	71.63	67.25	69.37	53.11	97.61
STANet	79.37	85.50	82.32	69.95	98.52
SNUNet	85.60	81.49	83.50	71.67	98.71
IFNet	96.91	73.19	83.40	71.52	98.83
ChangeFormer	91.83	88.02	89.88	81.63	99.12
BiFA	95.15	93.60	94.37	89.34	99.56
RSM-CD	93.37	90.42	91.87	84.96	–
SChanger	94.62	91.83	93.20	87.27	–
CDMamba	95.58	92.01	93.76	88.26	99.51
Mamba-CD	96.52	93.91	95.20	90.83	99.62
MambaRefine-CD (ours)	95.79	94.90	<u>95.34</u>	<u>91.10</u>	<i>99.56</i>
MambaRefine-CD VMamba	<u>96.32</u>	<u>94.68</u>	95.49	91.37	<u>99.57</u>

Table 4: Backbone scaling analysis on DSIFN-CD.

Backbone	Params (M)	GFLOPs	F1	IoU	OA
MambaVision-T	46.80	50.05	94.58	89.72	96.24
MambaVision-S	65.40	64.08	<u>95.67</u>	<u>91.71</u>	<u>96.98</u>
MambaVision-B	113.44	109.15	95.77	91.89	97.07

Table 5: Temporal-input ablation study on WHU-CD using VMamba-Tiny encoder.

ID	Temporal Input	Pre	Rec	F1	IoU	OA
T0	features_only	96.44	94.43	95.43	91.25	99.56
T1	signed_only	96.43	94.24	95.32	91.07	99.55
T2	abs_only	95.99	94.67	95.32	91.06	99.55
T3	features_signed	96.14	94.62	95.37	91.15	99.56
T4	features_abs	96.99	94.78	95.87	92.07	99.61
T5	signed_abs	96.02	94.75	95.38	91.17	99.56
T6	full	95.59	95.29	95.44	91.27	99.56

Table 6: Group-A temporal-difference ablation on DSIFN-CD using MambaVision-small.

ID	Temporal Mode	Pre	Rec	F1	IoU	OA	Thr.	Params (M)
A0	abs_only	94.49	94.82	<u>94.66</u>	<u>89.85</u>	<u>96.29</u>	0.60	46.43
A1	signed_only	94.21	95.23	94.72	89.97	96.32	0.60	46.43
A2	abs_signed	94.24	94.94	<i>94.59</i>	<i>89.73</i>	<i>96.24</i>	0.60	46.73

ID	$\Delta F1$ vs A2 (pp)	ΔIoU vs A2 (pp)
A0 abs_only	+0.07	+0.12
A1 signed_only	+0.13	+0.24
A2 abs_signed	0.00	0.00